

DH Part 3

EGR 445/545

GVSU

Plan for today

- DH review
- DH examples
- PUMA DH example
- Fanuc DH activity

Reminders

- check deadlines from google sheet

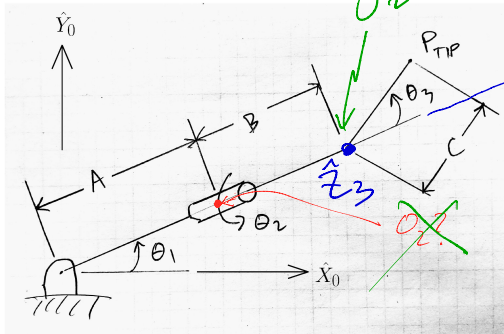
DH Review

Quick Poll

`https://etc.ch/NcCm`

DH Review Question

Which of the following is true of O_2 ?



What determines the location of O_2 ? $\hat{z}_2$ and $\hat{z}_3$
→ relationship between $\hat{z}_2$ and $\hat{z}_3$

- we have freedom regarding the location of O_2
- O_2 can go in the center of the joint
- O_2 must go in the center of the joint
- O_2 cannot go in the center of the joint

DH procedure review

- what are the two main phases of the DH procedure?

- phase 1: establish $\hat{z}$ axes

- phase 2: assign origins and $\hat{x}$ axes!

- where do most student mistakes happen?

→ phase 2

- assuming freedom in origin

locations

- $\hat{x}_{i-1}$ can never be $\parallel$ to $\hat{z}_i$

Key Parts of Chapter 3 - restated

- If you understand how **eqn. 3.6** is derived based on **Figure 3.15** and you understand the **parameter definitions** and **frame attachment** summaries from **section 3.4**, the rest of chapter 3 falls into place.

$${}^{i-1}_i T = HT_x(\alpha_{i-1}, \overset{x}{\underset{||}{\theta_{i-1}}}) \cdot HT_z(\theta_i, z = d_i)$$

Review: the Denavit-Hartenberg Convention

- what does the convention do for us?
- why do we need a convention?

But I want to be unconventional...

- what can go wrong if we don't use a convention?
 - [we have to formulate our own HT matrices]
 - lots of room for sign errors and other small mistakes
 - it is much harder to collaborate with others
 - it is harder for me to help you debug
 - and it makes grading much harder

Benefits of the DH convention

- we can use eqn 3.6 for our HT matrices
 - `my_robotics.DH` function uses eqn. 3.6
- it is easier to collaborate
- I can help you debug your issues faster and easier
- it really isn't that hard
- it isn't optional
 - yes, will be tested on this (repeatedly)

What does the DH convention actually do?

- tells us where to put our frames
 - origin location and unit vector orientation
- tells us how to move from one frame to the next
- models each robot link using four parameters: α , a , θ_i , and d

Key DH Questions

- where do the Z axes go?
- where do the X axes go?
- where do the origins go?

Phase 1

(steps 1+3)

Phase 2

(steps 2+4)

from post
↑

DH Principles

- Z_i axes along joint axes
 - rotational joints rotate around Z_i
 - prismatic joints translate along Z_i
 - X_{i-1} axis points along common perpendicular between Z_{i-1} and Z_i or perpendicular to Z_{i-1} and Z_i if they intersect
 - you must be able to rotate about X_{i-1} to get from Z_{i-1} to something parallel to Z_i
- X* • to get from frame $i - 1$ to frame i , follow these four steps in order:
1. rotate by α_{i-1} about X_{i-1}
 2. translate by a_{i-1} along X_{i-1}
 3. rotate by θ_i about Z_i
 4. translate by d_i along Z_i
- OTT*

DH Big Idea

The frames must be placed so that for each link it is possible to move from frame $i - 1$ to frame i by following the four steps in order:

1. rotate by α_{i-1} about X_{i-1}
2. translate by a_{i-1} along X_{i-1}
3. rotate by θ_i about Z_i
4. translate by d_i along Z_i

- This seems really simple, but 90%+ of student mistakes related to DH can be found by checking the four steps in order.

DH Frame Attachment Summary

1. Identify the joint axes
2. Identify the common perpendiculars or the point of intersection between the joint axes
3. Assign the Z_i axes along the i^{th} joint axes
4. Assign the X_i axes along the common perpendiculars or perpendicular to both Z axes if two consecutive Z_i intersect
5. Assign the Y axes to complete a right-hand coordinate system
 - normally skipped
6. Decide on frames $\{0\}$ and $\{N\}$

Simplified DH Procedure

- identify joint axes and assign Z_i axes
- determine Z axis case based on the relationship between Z_{i-1} and Z_i
 - case 1: parallel
 - case 2: intersecting
 - case 3: non-parallel and non-intersecting
- locate origins and assign X_{i-1} axes based on Z axis case
- assign frames 0 and N

Two Main Phases to the DH Convention

1. assign the $\hat{Z}$ axes
2. assign the origins and the $\hat{X}$ axes
 - the relationship between $\hat{Z}_i$ and $\hat{Z}_{i-1}$ determines where O_{i-1} goes and where $\hat{X}_{i-1}$ points

Three Cases for Z Axes

1. Z_{i-1} and Z_i are parallel
2. Z_{i-1} and Z_i intersect
3. Z_{i-1} and Z_i are not parallel and do not intersect

∞ common $\perp$'s, O_{i-1} along Z_{i-1} ; some freedom

→ O_{i-1} at the intersection; $\hat{X}_{i-1} \perp$ to plane formed by $\hat{Z}_{i-1}$ and $\hat{Z}_i$

→ only 1 common $\perp$; $\hat{X}_{i-1}$ along common $\perp$; O_{i-1} at intersection of common $\perp$ and $\hat{Z}_{i-1}$

Case 1: Parallel Z Axes

- this is the only case where you have *some* freedom in placing an origin
 - but your freedom is limited to sliding the origin along Z_{i-1}
- X_{i-1} still goes along the common perpendicular line between Z_i and Z_{i-1}
 - but now there are infinitely many common perpendiculars

Case 2: Intersecting Z Axes

- if two consecutive Z axes intersect:
 - the origin for frame $i - 1$ must go at the intersection
 - X_{i-1} must be perpendicular to the plane formed by Z_{i-1} and Z_i
- **why?**

Case 3: Non-parallel and Non-intersecting

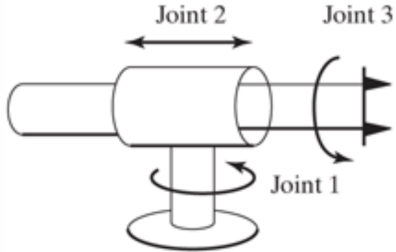
- there is only one common perpendicular line
- X_{i-1} must go along the common perpendicular
- the origin for frame $i - 1$ must go at the intersection of the common perpendicular and Z_{i-1}
 - i.e. X_{i-1} and Z_{i-1} intersect at the origin of frame $i - 1$
- this case is shown in Fig. 3.15

DH Examples

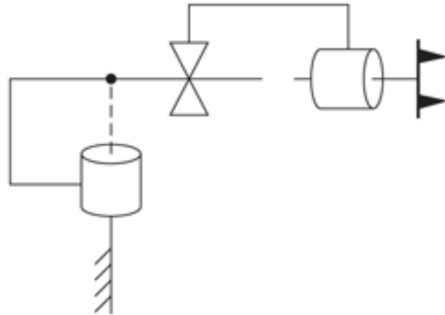
Example 3.4

- page 75

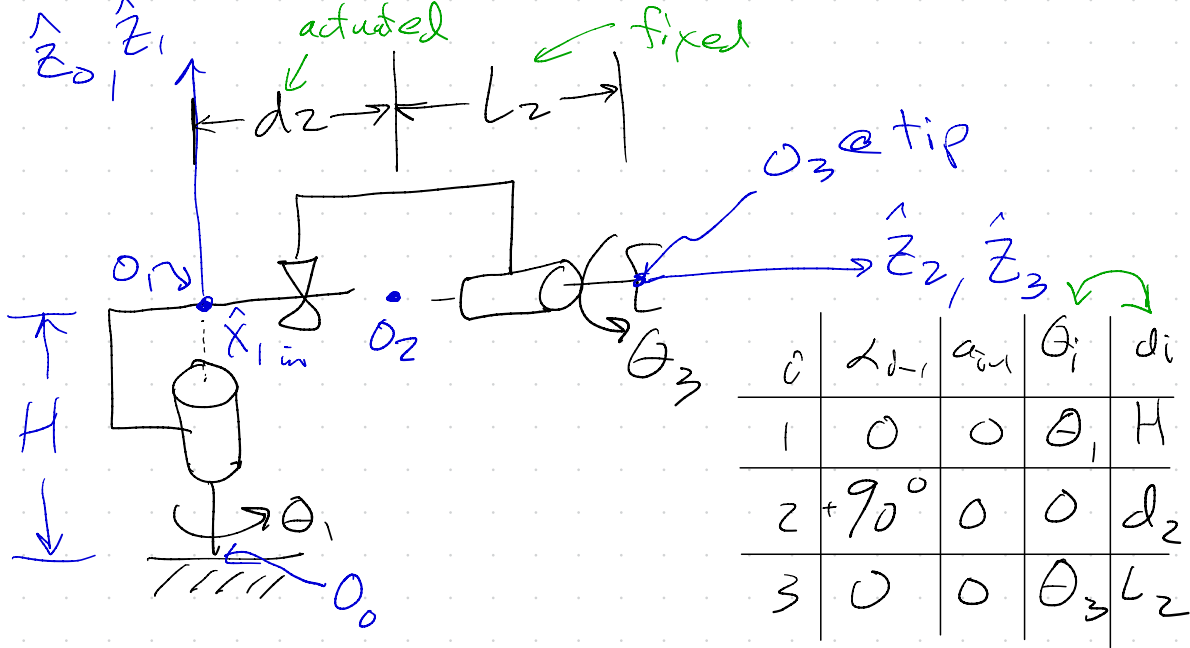
FIGURE 3.9: Manipulator having three degrees of freedom and one prismatic joint.



(a)



(b)

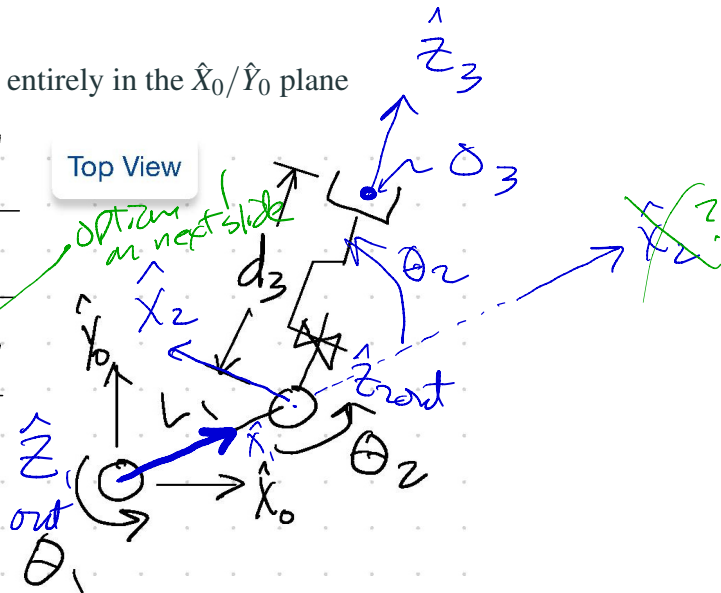


Modified SCARA 1: RRP Planar

Note: The robot moves entirely in the $\hat{X}_0/\hat{Y}_0$ plane

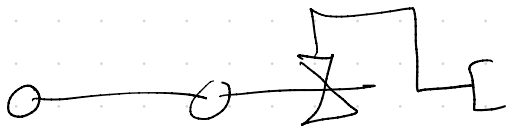
i	α	a	θ	d
1	0	0	θ_1	0
2	0	L_1	$\theta_2 + 90^\circ$	0
3	90°	0	0	d_3

Top View



Planar SCARA home position question

Option 1



$$\theta_2 = 0$$

→ need $\theta_2 + 90$
in table

option 2

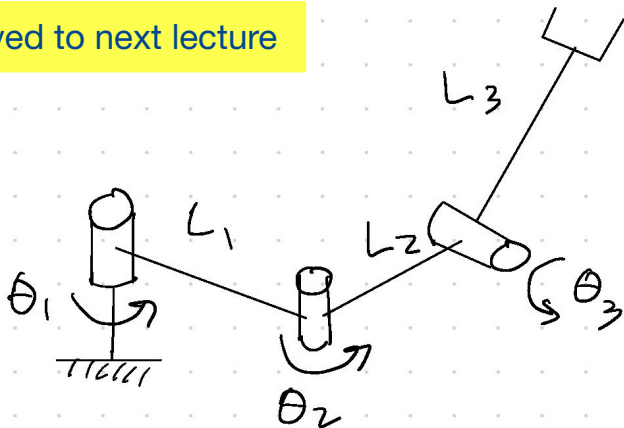


row 2 only needs θ_2
(no $+90$ term)

Modified SCARA 2: 3R with Rotated Third Joint

Note: L_1 and L_2 move in the $\hat{X}_1/\hat{Y}_1$ plane and $\hat{Z}_3$ is in that same plane

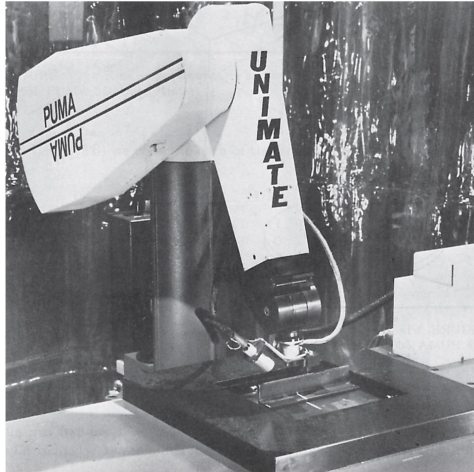
Moved to next lecture



PUMA 560

- section 3.7, starting on page 83

Moved to next lecture



Courtesy of Unimation Incorporated, Shelter Rock Lane, Danbury, Conn.

Fanuc DH Activity

Fanuc DH Group Activity

- work in groups of 2-3
- decide whether you are working on the yellow or white Fanuc
- draw a schematic of the robot
- apply the DH procedure to the robot
- fill out a table
- be prepared to discuss your work

Yellow Fanuc Schematic

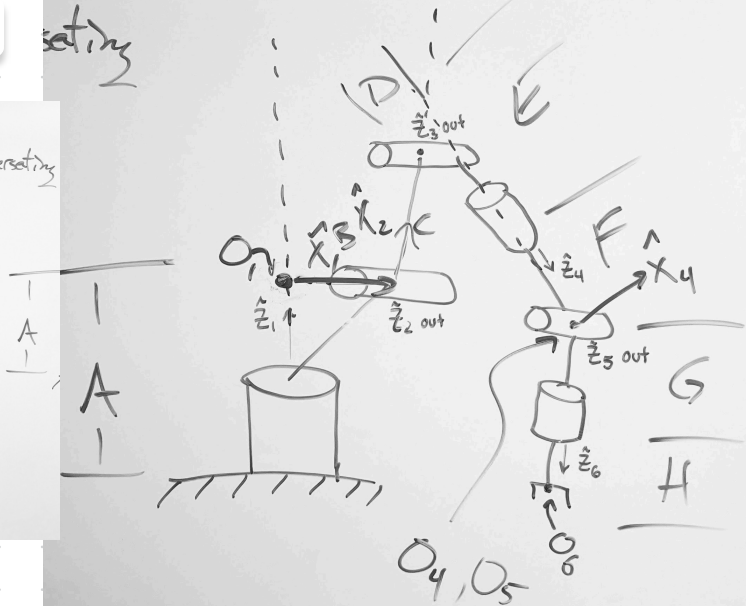
$\hat{z}_1$ and $\hat{z}_2 \rightarrow$ case 3
non-parallel / non-intersecting

$\hat{z}_2 \parallel \hat{z}_3$

$\hat{z}_3$ and $\hat{z}_4$: case 3 non/non

$\hat{z}_4$ intersects $\hat{z}_5$: case 2

$\hat{z}_5$ and $\hat{z}_6$ also intersect



Yellow Fanuc DH table and home position

i	α_{i-1}	a_{i-1}	θ_i	d_i
1	0	0	θ_1	A
2	$+90$	B	θ_2	0
3	0	C	θ_3	0
4	$+90$	D	θ_4	E + F
5	-90	0	θ_5	0
6	$+90$	0	θ_6	G + H

Home Position without Joint offsets
 $\theta_i = 0$

